SCX Causal Unidentifiability Theorem

SCX

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Abstract

Proves that causal direction claims are unidentifiable from observational data without declared identification assumptions. Three mutually incompatible Structural Causal Models produce identical observational distributions.

0.1 Causal Counterfactual Corollary

The same identification barrier appears in ordinary causal claims. A claim such as "X causes Y" is not a statement about the observed event alone: it asserts a counterfactual contrast, e.g. that Y would have been different under an intervention $\mathrm{do}(X = x')$. The counterfactual world is not jointly observed with the factual world. Hence the claim has empirical content only after the analyst declares an identification strategy.

Theorem 0.1 (SCX Causal Unidentifiability Theorem)**.** *Let (X, Y, Z) take values in standard Borel spaces and let P_{XYZ} be any observed joint law admitting regular conditional distributions. Suppose the target causal claim is a counterfactual statement about the effect of X on Y, for example $P(Y \in B \mid \mathrm{do}(X = x), Z = z)$ or an average treatment effect $\psi = \mathbb{E}[Y_x - Y_{x'}]$. Without a declared identification assumption, the observational law P_{XYZ} does not identify the causal direction between X and Y. In particular, there exist structural causal models inducing the same observed joint law P_{XYZ} with the following mutually incompatible causal interpretations:*

(a) $X \to Y$: the association is generated by a direct causal effect of X on Y;

(b) $C \to X$ and $C \to Y$: the association is generated by a common cause C whose observed component is Z;

(c) $Y \to X$: the association is generated by reverse causation.

If $P_{Y|X,Z}$ or $P_{X|Y,Z}$ varies in the conditioning argument, the corresponding directed model has an active controlled effect. If the observed association is only marginal, the same construction applies after dropping Z from the conditioning set and generating Z from $P_{Z|X,Y}$. Consequently no statistic of observational samples from P_{XYZ} can verify which of these counterfactual interpretations is true.

Proof. **[Rigor: Full Proof]**The proof is constructive. Let U_Z, U_X, U_Y be independent $\mathrm{Unif}(0, 1)$ random variables, and let $Q_{A|B}$ denote a measurable conditional quantile or, more generally, a randomization kernel that generates the conditional law $P(A \mid B)$.

World A: direct causation $X \to Y$. Define the SCM

$$\begin{aligned} Z_A &= Q_Z(U_Z), \\ X_A &= Q_{X|Z}(U_X \mid Z_A), \\ Y_A &= Q_{Y|X,Z}(U_Y \mid X_A, Z_A). \end{aligned} \tag{1}$$

Its induced observational law is

$$P_A(dz, dx, dy) = P_Z(dz)\, P_{X|Z}(dx \mid z)\, P_{Y|X,Z}(dy \mid x, z) = P_{XYZ}(dz, dx, dy), \tag{2}$$

by the chain rule for probability kernels. When $P_{Y|X,Z}$ varies with x, this model has a nonzero controlled causal effect of X on Y conditional on Z.

World B: common-cause confounding. For each z, choose a latent variable $U_C \sim \mathrm{Unif}(0, 1)$ and measurable functions g_X, g_Y such that

$$(g_X(z, U_C), g_Y(z, U_C)) \sim P_{XY|Z=z}. \tag{3}$$

This is possible by the randomization lemma for regular conditional distributions. Define

$$\begin{aligned} Z_B &= Q_Z(U_Z), \\ C_B &= (Z_B, U_C), \\ X_B &= g_X(Z_B, U_C), \\ Y_B &= g_Y(Z_B, U_C). \end{aligned} \tag{4}$$

Then X_B and Y_B have no directed edge between them; their dependence is generated by the common cause C_B, whose observed component is Z_B. The induced law is

$$P_B(dz, dx, dy) = P_Z(dz)\, P_{XY|Z}(dx, dy \mid z) = P_{XYZ}(dz, dx, dy). \tag{5}$$

If the analyst additionally declares that the observed Z is the complete common cause and that the exogenous errors are independent, this construction specializes to $Z \to X$ and $Z \to Y$ exactly when $X \perp\!\!\!\perp Y \mid Z$. Without that completeness declaration, the latent component U_C is observationally invisible.

World C: reverse causation $Y \to X$. Define the SCM

$$\begin{aligned} Z_C &= Q_Z(U_Z), \\ Y_C &= Q_{Y|Z}(U_Y \mid Z_C), \\ X_C &= Q_{X|Y,Z}(U_X \mid Y_C, Z_C). \end{aligned} \tag{6}$$

Again, by the chain rule,

$$P_C(dz, dx, dy) = P_Z(dz)\, P_{Y|Z}(dy \mid z)\, P_{X|Y,Z}(dx \mid y, z) = P_{XYZ}(dz, dx, dy). \tag{7}$$

Equations (**??**), (**??**), and (**??**) show that the three SCMs induce identical observational distributions over (X, Y, Z) while assigning different counterfactual meanings to the same association. Let $T_n = T((X_i, Y_i, Z_i)_{i=1}^n)$ be any statistic or verification procedure based only on observational samples. Since the sample law is $P_{XYZ}^{\otimes n}$ in all three worlds, the distribution of T_n is identical in all three worlds. Therefore T_n cannot verify that World A, World B, or World C is the true counterfactual data-generating process. The missing information is precisely the unobserved counterfactual response under interventions. □

Corollary 0.2 (No-Assumptions Causal Claims Have Zero Audit Content)**.** *Let $\mathcal{A}_{\mathrm{id}}$ be the set of declared identification strategies, including randomized assignment, a valid backdoor adjustment set, a valid front-door mediator, a valid instrumental variable, regression discontinuity, or another explicit design that identifies $P(Y \mid \mathrm{do}(X))$. A causal claim about X and Y that declares no element of $\mathcal{A}_{\mathrm{id}}$ is epistemically equivalent, in the SCX sense, to reporting the observational law P_{XYZ} alone.*

Proof. By Theorem **??**, the equivalence class of SCMs compatible with P_{XYZ} contains at least one direct-causal, one common-cause, and one reverse-causal interpretation. A declared identification strategy is exactly an assumption that removes some members of this equivalence class. If no such strategy is declared, the identified set for the causal estimand contains all counterfactual values generated by the compatible SCMs. The claim therefore does not shrink the equivalence class beyond what was already implied by P_{XYZ}. This is the causal instance of Theorem **??**: without declared invariance or identification assumptions, the source of the observed association is unidentifiable. $\square$